

A Knudsen-extended, calendering-aware Zehner–Bauer–Schlünder closure for the effective thermal conductivity of lithium-ion electrode coatings and separators

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Abstract

The effective through-plane thermal conductivity λ_{eff} of electrode coatings and separators is a controlling input of lithium-ion battery thermal models, yet it is usually parameterized by porosity-only closures with continuum pore-gas conductivity and no uncertainty statement. We develop and test a closure that repairs three specific deficiencies. (i) *Pore-scale gas physics*: a systematic Smoluchowski/Knudsen correction for micrometer-scale electrode pores (3–6% in the helium atmosphere of laser-flash metrology), decisive for polyolefin separators, whose 43–64 nm pores ($\text{Kn} = 1.0\text{--}1.6$) lose 84% of their pore-gas conduction. (ii) *Process microstructure*: a compression-dependent contact term $\varphi(\Pi)$ whose functional form encodes published adhesion-strength observations. (iii) *Inverse use*: an end-to-end differentiable implementation that converts thermal metrology into manufacturing quality control. Against the open calendering dataset of Gandert et al. (4 electrode types $\times$ 27 states), the zero-fit model predicts the electrode closest to its assumptions (uncalendered NMC811) to +2.7%, outperforms classical effective-medium baselines (Wiener bounds, Maxwell–Eucken, Bruggeman), and an ablation shows mean error falling $31.1\% \rightarrow 13.5\% \rightarrow 4.5\%$ as the contact physics is added — reaching the measurement noise floor and reproducing, for the first time to our knowledge, the experimentally observed u-shaped conductivity–calendering dependence. A profile-likelihood identifiability analysis and a hybrid-swap decomposition of cross-recipe transfer failure show that a single re-anchorable parameter — the as-coated bridge conductance, set by the conductive-additive recipe — carries the transfer error, yielding material design rules. The differentiable closure recovers coating porosity from one 3%-noise thermal measurement to ± 0.008 absolute and provides the first candidate inline monitor for interface/delamination quality ($z = 5\text{--}22$). We specify the validation experiments, including a two-gas-pressure protocol that separates pore-gas from skeleton conduction.

Keywords: effective thermal conductivity; lithium-ion battery; calendering; Knudsen effect; effective-medium theory; inline quality control

1 Introduction

Heat must leave a lithium-ion cell through its layered stack, and the through-plane effective thermal conductivity of the porous electrode and separator is the bottleneck: cell-level cross-plane values of $0.15\text{--}0.8 \text{ W m}^{-1} \text{ K}^{-1}$ lie two orders of magnitude below in-plane values, and through-plane temperature gradients accelerate aging [?, ?, ?]. Battery thermal models therefore need λ_{eff} of every layer as input, and they need it *as a function of manufacturing parameters*: calendering sets the electrode porosity and changes λ_{eff} by tens of percent across a normal process window [?].

The porous-media community provides closures for exactly this quantity, and the Zehner–Bauer–Schlünder (ZBS) family [?, ?, ?] is the most widely used, including for battery components [?, ?]. Yet recent measurements expose a structural mismatch between what these closures model and what electrode manufacturing produces. Gandert et al. [?] measured four electrode types across 27 calendering states and found a *non-monotonic, u-shaped* dependence of λ_{eff} on porosity that correlates with coating/collector adhesion; they compared two state-of-the-art models — an analytic homogenization closure [?] and a discrete-element simulation [?]. They reported poor agreement: both models rise monotonically as porosity falls. The same authors flag a second unresolved issue: the gas in micrometer-scale pores

is not a continuum, and the resulting Knudsen-type reduction of pore-gas conduction is "a known but unquantified correction". Finally, closures are used almost exclusively in the forward direction; their potential as *inverse* tools — reading porosity or interface quality back from a fast thermal measurement — is unexploited, although interface quality dominates stack thermal resistance [?] and has no inline monitor today.

This paper addresses these gaps with a deliberately minimal extension of the ZBS closure, validated and calibrated entirely on open published data. Our contributions are:

- C1 A Knudsen-extended ZBS closure** for battery components (Sec. ??). We quantify the correction the literature flags but does not compute: 3–6% for electrodes under laser-flash conditions (helium), –84% of pore-gas conduction for polyolefin separators, where we further show that classical continuum sphere-pack models match measurements only through a cancellation of two errors (Sec. ??).
- C2 A calendering-aware contact term** $\varphi(\Pi) = \varphi_0 + a\Pi + b\Pi^2$ (Sec. ??), the first closure to our knowledge that reproduces the measured u-shape. An ablation (Sec. ??) attributes the error reduction 31.1% $\rightarrow$ 13.5% $\rightarrow$ 4.5% to its components; the endpoint is the dataset’s own noise floor.
- C3 An error, identifiability and transfer analysis** (Sec. ??): profile likelihood bounds the λ_s/φ_0 confound; hybrid-swap experiments show cross-recipe transfer failure is carried by one parameter (the as-coated bridge conductance), giving a one-measurement re-anchoring rule and volume-space design principles for additive and binder selection (Sec. ??).
- C4 Differentiable inverse tools for manufacturing QC** (Sec. ??): porosity from a single 3%-noise thermal reading to ± 0.008 absolute with calibrated uncertainty, and a bridge/core decomposition that makes thermal metrology the first candidate inline monitor for interface/delamination quality.
- C5 A falsifiable experimental program** (Sec. ??), centered on a two-gas-pressure protocol that separates pressure-dependent pore-gas conduction from pressure-independent skeleton conduction, testing the mechanism rather than the fit.

All claims trace to executed, openly available code; data transcriptions carry full citations (Reproducibility Statement).

2 Related work

Analytic effective-medium closures. The classical hierarchy brackets two-phase conduction between the Wiener series/parallel bounds [?], tightened by Hashin–Shtrikman [?]; structure-specific models then commit to a morphology: Maxwell–Eucken [?, ?] assumes one phase dispersed in the other, Bruggeman’s symmetric effective-medium theory [?] assumes co-continuity, and the ZBS family [?, ?, ?] resolves a packed-bed unit cell with point (or, in Bauer’s extension, flattened) particle contacts. These models are instantaneous to evaluate, differentiable, and physically interpretable, but their fidelity is hostage to the assumed morphology — a property we exploit diagnostically in Sec. ??, where the *pattern* of baseline failures (Bruggeman matching co-continuous separators but failing sphere-packed electrodes by up to +465%, and vice versa for ZBS) identifies the governing microstructure. None of these closures, in their standard battery usage, includes rarefied-gas corrections or process variables beyond porosity.

Microstructure-resolved simulation. Discrete-element models [?] and image-based or virtual-microstructure homogenization [?] resolve particle arrangements explicitly and can, in principle, represent contact areas. In practice both approaches reduced their predictions to porosity-parameterized curves when compared to the calendering data, and both failed to reproduce the measured u-shape [?]: the missing physics (calender-induced contact damage and recovery at the coating/collector interface and between particles) was not represented in either microstructure generator. Resolved simulation also costs orders of magnitude more per evaluation than a closure and is not differentiable for inverse use. Our approach is complementary: we keep closure-level cost and differentiability and inject the missing contact physics through a low-dimensional, process-indexed term whose parameters are interpretable and cheaply calibrated (≈ 6 calendering states).

Rarefied gas conduction in confined pores. The pressure- and size-dependent reduction of gas conduction (temperature-jump/Smoluchowski effect) is textbook kinetic theory [?] and quantitatively central in insulation and aerogel research [?], where pore sizes resemble battery separators. Its coupling into the

ZBS framework for battery components appears to be missing from the literature; Gandert et al. [?] explicitly note the gap. Sec. ?? supplies the term, and Sec. ?? the experiment that isolates it (gas conduction scales with pressure; skeleton conduction does not).

Battery component thermal measurements. Component-level data come from steady-state stacked-sample rigs with controlled compaction [?, ?], copper-block reference-bar setups on cell teardowns [?, ?], and flash diffusivity with penetration-model evaluation [?, ?]; early canonical values are due to Maleki et al. [?]. Two robust findings frame our work: electrolyte soaking raises coating conductivity by roughly threefold [?, ?], and interfaces, not bulk layers, dominate stack resistance [?] — which both motivates the contact term and warns that stack-derived values are lower bounds for single coatings.

3 Model

3.1 Setting, notation, and assumptions

We model a porous coating as a two-phase medium: solid particles (conductivity λ_s , diameter d_p) and a pore fluid (λ_f ; gas or electrolyte) at porosity ψ , seeking the effective through-plane conductivity $\lambda_{\text{eff}} = k_{\text{bed}} \lambda_f$. The compression rate of calendering is $\Pi = 1 - s_{\text{co}}/s_{\text{co},0}$, with s_{co} the coating thickness. Assumptions:

- A1** Steady diffusive conduction; radiation and pore-scale convection negligible (verified: at coating scale, $\text{Ra}^* \sim 10^{-15}$; radiation negligible at μm pores and ambient T).
- A2** Statistically isotropic sphere-pack morphology for the closure’s unit cell; deviations (graphite flakes, co-continuous separator skeletons) are not re-meshed but absorbed interpretably — flake anisotropy into an effective λ_s bounded by the single-crystal anisotropy band, contact/bridge conduction into the term of Sec. ??.
- A3** The pore gas behaves as a continuum except for the conductivity reduction of Eq. (??); liquid electrolyte needs no correction.
- A4** One scalar φ represents the area fraction of all solid–solid conduction bridges (particle–particle and coating–collector); its conductivity λ_b is that of the bridging species.
- A5** Within one electrode family (one recipe and coating process), $(\lambda_s, \lambda_b, \varphi_0, a, b)$ are constants; only ψ and Π vary along the calendering trajectory.

The base closure was validated in companion work on Zehner’s original packed-bed data across $\kappa = \lambda_s/\lambda_f \in [4, 14094]$; the battery systems’ κ values (14–1005) lie strictly inside this range, so the closure *interpolates* in κ and extrapolates only in ψ (electrode porosities 0.20–0.61 vs. packed-bed 0.37–0.43); this extrapolation is examined empirically in Sec. ??.

3.2 Base closure (M0)

The ZBS closure [?, ?, ?] writes the bed conductivity ratio as a parallel arrangement of a fluid path and a particle-core path,

$$k_{\text{bed}}(\psi, \kappa) = 1 - \sqrt{1 - \psi} + \sqrt{1 - \psi} \lambda_{pm}(\psi, \kappa), \quad (1)$$

with the unit-cell core term

$$\lambda_{pm} = \frac{2}{1 - B/\kappa} \left[\frac{B(\kappa - 1)}{\kappa(1 - B/\kappa)^2} \ln \frac{\kappa}{B} - \frac{B + 1}{2} - \frac{B - 1}{1 - B/\kappa} \right], \quad B = C \left(\frac{1 - \psi}{\psi} \right)^{10/9}, \quad (2)$$

and shape factor $C = 1.25$ (spheres). Equations (??)–(??) with continuum λ_f constitute baseline M0.

3.3 Knudsen-corrected pore gas (M1 fluid model)

When the molecular mean free path Λ approaches the pore size, wall collisions throttle gas heat transport. We use the Smoluchowski form [?, ?]

$$\lambda_{g,\text{eff}} = \frac{\lambda_g}{1 + 2\beta \text{Kn}}, \quad \text{Kn} = \frac{\Lambda}{d_{\text{pore}}}, \quad \Lambda = \frac{k_B T}{\sqrt{2} \pi d_g^2 p}, \quad (3)$$

with $\beta = 1.64$ for air on technical surfaces (β -sensitivity quantified in Sec. ??; d_g the molecular collision diameter, so $\Lambda_{\text{air}} \approx 67$ nm and $\Lambda_{\text{He}} \approx 190$ nm at ambient conditions). The pore size follows the hydraulic estimate for sphere packings,

$$d_{\text{pore}} = \frac{2}{3} \frac{\psi}{1-\psi} d_p, \quad (4)$$

or is taken from porosimetry where available (separators: 43–64 nm). The corrected $\kappa_{\text{eff}} = \lambda_s/\lambda_{g,\text{eff}}$ enters Eq. (??) unchanged.

3.4 Calendering-aware contact term (M2)

Calendering acts on contact zones, not on pore volume — which is why porosity-only models cannot represent its signature. We adopt the VDI flattened-contact generalization of Eq. (??) [?], extended with a bridge phase whose conductivity may differ from the particles’:

$$k_{\text{bed}} = 1 - \sqrt{1-\psi} + \sqrt{1-\psi} \left[\varphi \frac{\lambda_b}{\lambda_f} + (1-\varphi) \lambda_{pm}(\psi, \kappa) \right], \quad (5)$$

with φ the bridge area fraction; $\lambda_b \approx 130$ W m⁻¹ K⁻¹ for graphite–graphite flake contacts, $\lambda_b \approx 24$ W m⁻¹ K⁻¹ for the carbon-black networks bridging NMC particles [?]. Setting $\varphi = 0$ recovers M0/M1 exactly. Guided by the adhesion-strength phenomenology reported with the calendering data [?] — interfacial shear damage at moderate line loads, particle interlocking and foil penetration at high loads — we parameterize

$$\varphi(\Pi) = \max(0, \varphi_0 + a\Pi + b\Pi^2), \quad (6)$$

expecting $\varphi_0 > 0$ (as-coated bridge network), $a < 0$ (damage), $b \geq 0$ (recovery; a threshold effect that may not be reached within a process window). M1 denotes Eqs. (??)–(??) restricted to constant φ ($a = b = 0$); M2 the full quadratic.

3.5 Differentiable implementation and inverse use

Equations (??)–(??) are implemented in JAX (64-bit), so $\partial\lambda_{\text{eff}}/\partial\psi$ and all calibration gradients are exact. The inverse map for porosity QC is Newton’s iteration with autodiff derivatives,

$$\psi_{k+1} = \psi_k - \frac{\lambda_{\text{eff}}(\psi_k) - \lambda^{\text{meas}}}{\partial\lambda_{\text{eff}}/\partial\psi(\psi_k)}, \quad (7)$$

unique because $\lambda_{\text{eff}}(\psi)$ is strictly monotonic over the relevant range (verified numerically), and accurate to machine precision in round-trip tests. First-order uncertainty propagation gives

$$\sigma_\psi = \frac{\sqrt{\sigma_{\text{meas}}^2 + \sigma_{\text{model}}^2} \lambda_{\text{eff}}}{|\partial\lambda_{\text{eff}}/\partial\psi|}, \quad (8)$$

with σ_{meas} the relative metrology noise and σ_{model} a model-form floor (2%, the residual scatter of the packed-bed validation).

4 Data, baselines, and calibration protocol

4.1 Primary dataset

Gandert et al. [?] (open access, CC-BY): two graphite anodes (Cu foil; 96 wt% active material, CMC/SBR binder, 1.5% carbon black) and NMC622/NMC811 cathodes (Al foil; PVDF binder; NMC622 additionally 2 wt% flake-graphite additive), calendered on a laboratory roll calender to 6–8 states each. λ_{eff} follows from $\lambda = \kappa_d \rho c_p$ with laser-flash diffusivity under *helium* purge (penetration-model evaluation [?]), DSC heat capacity, pycnometric solid density, and flat-anvil micrometer thickness (≤ 0.3 N mm⁻²; ball-tip gauges compress the coating and read up to 23% low [?]); porosity uncertainties follow GUM. We transcribe per-sheet porosities/thicknesses and evaluate the published quadratic fits at those porosities; the foil is removed in series,

$$\frac{s_{\text{co}}}{\lambda_{\text{co}}} = \frac{s_{\text{stack}}}{\lambda_{\text{stack}}} - \frac{s_{\text{cc}}}{\lambda_{\text{cc}}}, \quad (9)$$

($\lambda_{\text{Cu}} = 400$, $\lambda_{\text{Al}} = 237$ W m⁻¹ K⁻¹; correction $< 0.1\%$); the coating/collector contact resistance remains lumped in λ_{co} , as in the source. All dry comparisons use helium as pore gas ($\lambda_g = 0.1518$ W m⁻¹ K⁻¹) — with $\Lambda_{\text{He}} \approx 190$ nm this triples the Knudsen term relative to air and makes the dataset an unusually sharp test of Eq. (??).

4.2 Anchor datasets

Dry polyolefin separators $0.07\text{--}0.18\text{ W m}^{-1}\text{ K}^{-1}$ across types and compaction [?]; ceramic-coated separator stack 0.10 dry / 0.11 wet [?]; soaked/dry pairs: graphite $0.30 \rightarrow 0.89$, LCO $0.36 \rightarrow 1.10$ [?], CMS graphite $0.57 \rightarrow 1.35$, LMO $0.16 \rightarrow 0.45\text{ W m}^{-1}\text{ K}^{-1}$ [?]; interface dominance from [?]; separator pore sizes from porosimetry. Constituent properties: $\lambda_{\text{air}} = 0.026$, $\lambda_{\text{electrolyte}} = 0.18$ (EC/DMC class [?]), $d_p = 18\text{ }\mu\text{m}$ (graphite), $10\text{ }\mu\text{m}$ (NMC).

4.3 Baselines

We compare against the classical closure hierarchy, all evaluated with the *same* constituent properties and the same Knudsen-corrected pore gas (so differences isolate morphology assumptions, not inputs): Wiener series/parallel bounds [?]; Maxwell–Eucken with continuous fluid [?, ?],

$$\lambda_{\text{eff}}^{\text{ME}} = \lambda_f \frac{\lambda_s + 2\lambda_f + 2(1-\psi)(\lambda_s - \lambda_f)}{\lambda_s + 2\lambda_f - (1-\psi)(\lambda_s - \lambda_f)}; \quad (10)$$

Bruggeman’s symmetric EMT [?], the root of

$$\psi \frac{\lambda_f - \lambda_{\text{eff}}}{\lambda_f + 2\lambda_{\text{eff}}} + (1-\psi) \frac{\lambda_s - \lambda_{\text{eff}}}{\lambda_s + 2\lambda_{\text{eff}}} = 0; \quad (11)$$

and M0 (point-contact ZBS+Knudsen). The two microstructure-resolved literature models are not re-implemented; their failure on this dataset (monotonic in ψ , no u-shape) is documented by its authors [?].

4.4 Calibration, ablation, and error-analysis protocol

Per electrode family, parameters $\theta = (\lambda_s, \varphi_0, a, b)$ minimize relative residuals,

$$\hat{\theta} = \arg \min_{\theta \in \Theta} \sum_i \left[\frac{\lambda_{\text{eff}}(\psi_i, \Pi_i; \theta) - \lambda_{\text{co},i}^{\text{meas}}}{\lambda_{\text{co},i}^{\text{meas}}} \right]^2, \quad (12)$$

by trust-region least squares within physical bounds Θ : $\lambda_s \in [5, 139]\text{ W m}^{-1}\text{ K}^{-1}$ (graphite anisotropy band) or $[1.5, 5]$ (NMC); $\varphi_0 \in [0, 0.08]$; $a \in [-0.2, 0.3]$; $b \in [0, 0.8]$. The metric is the mean absolute percentage error (MAPE) over all sheets of a family. The protocol includes: (i) an *ablation* M0→M1→M2; (ii) *held-out transfer*: calibrate on graphite_{thin}, predict graphite_{thick} (same recipe, doubled coating thickness) and calibrate on NMC622, predict NMC811 (different additive recipe); (iii) *profile likelihood* over λ_s (refit remaining parameters on a grid) to bound identifiability; (iv) *hybrid swaps* to attribute transfer error to individual parameters.

5 Results and analysis

5.1 Knudsen magnitudes and zero-fit reference values

Figure ?? quantifies contribution C1. At reference states ($\psi = 0.30$ electrodes, 0.40 separator) the pore-gas reduction in air is 4.3% (graphite anode, $d_{\text{pore}} = 4.9\text{ }\mu\text{m}$), 8.8% (NMC, $2.3\text{ }\mu\text{m}$), and 84% for the separator at its measured 43–64 nm pores, where $\text{Kn} = 1.0\text{--}1.6$ — the mean free path *exceeds* the pore size. The effect on dry λ_{eff} is $-3.4\%/ -5.8\%$ for the electrodes; under the helium atmosphere of flash metrology these grow to 2.8–5.5% — comparable to the reproducibility of the measurement itself, hence worth carrying, and supplied here for the configuration in which the primary dataset was acquired. Zero-fit reference values (Fig. ??) land inside published ranges with no electrode-specific fitting: anode 2.56 (wet) / 0.60 (dry); cathode 0.97/0.31; separator 0.29/0.065 $\text{W m}^{-1}\text{ K}^{-1}$ (0.116 if the Knudsen term is wrongly omitted). Predicted soak ratios ($4.3\times$ anode, $3.2\times$ cathode) modestly exceed measured ones ($2.4\text{--}3.1\times$ [?, ?]) — the expected signature of a fluid-independent bridge backbone that M0 lacks, anticipating Sec. ?. Porosity sensitivity is large: a routine ± 0.02 calendaring tolerance moves electrode λ_{eff} by 5–9% (exact gradients), so treating λ_{eff} as a material constant in cell models embeds a porosity-correlated error of that size.

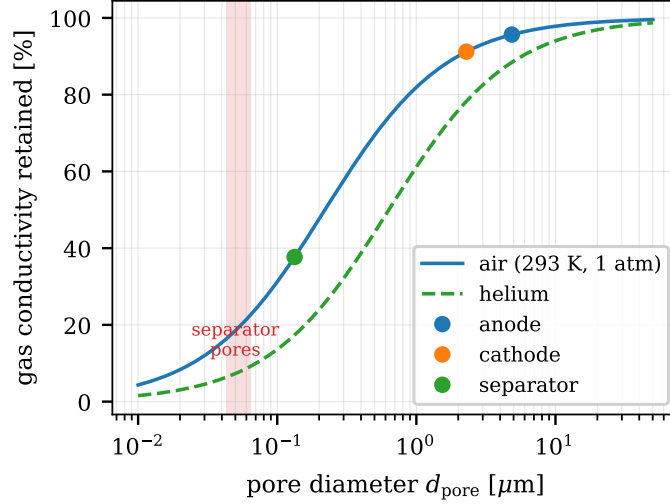


Figure 1: Retained fraction of pore-gas conduction, Eq. (??), at 293 K and 1 atm for air (solid) and helium (dashed). Markers: hydraulic pore sizes of the reference systems. Band: measured polyolefin-separator pore diameters — inside it, the air mean free path exceeds the pore size.

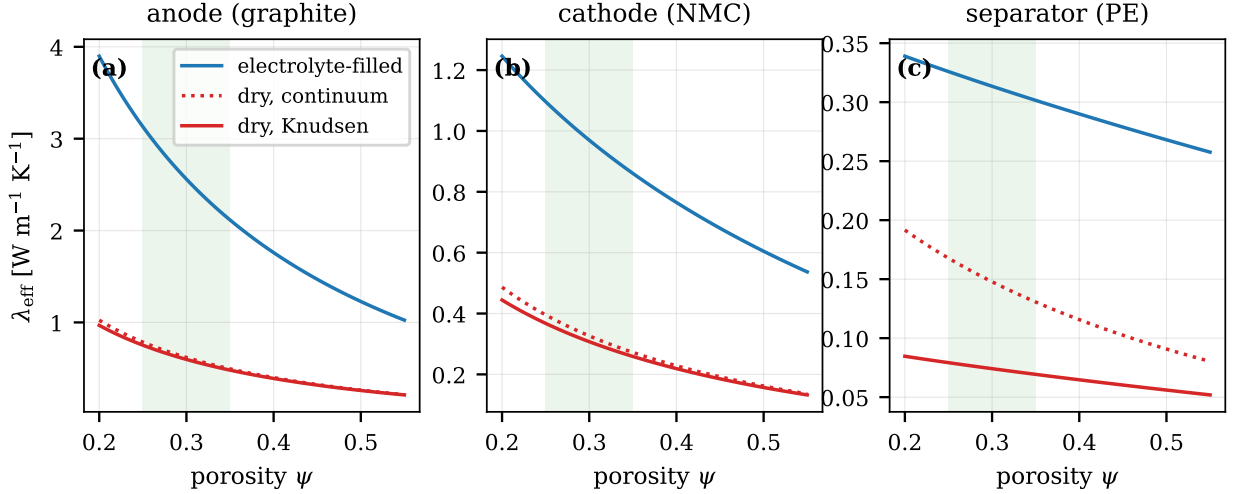


Figure 2: Zero-fit $\lambda_{\text{eff}}(\psi)$ for (a) graphite anode, (b) NMC cathode, (c) PE separator: electrolyte-filled, dry-continuum, and dry-Knudsen variants. Green band: typical post-calendering window.

5.2 Baseline comparison: morphology is the discriminator

Table ?? evaluates the closure hierarchy at the four uncalendered points — the cleanest test, since no calender-induced contact alteration exists there — using identical inputs and Knudsen-corrected helium pore gas throughout. The Wiener bounds are uninformative at graphite’s phase contrast (spanning 0.24–32 $\text{W m}^{-1} \text{K}^{-1}$). The structure-committed baselines fail in *opposite, morphology-diagnostic* directions: Bruggeman’s co-continuous EMT, which percolates the solid, overshoots by +17% (NMC622) to +465% (graphite); Maxwell–Eucken, which isolates each particle in fluid, undershoots by –11% to –70%. **M0** — spheres touching at points — is the only zero-parameter closure that is both well-ordered and accurate where its morphology assumption actually holds: +2.7% for NMC811, the electrode with the least additive (4 wt%) and hard, near-spherical particles. Its systematic *under*-prediction elsewhere (–29% NMC622, –36/–39% graphite) is ordered exactly by expected bridge strength (flake-graphite additive; plastically self-bridging soft flakes), and the residual walks monotonically from negative to positive along every calendering series: the u-shape is invisible to all porosity-only models, consistent with the documented failure of the two microstructure-resolved models [?]. The same diagnostic logic settles the separator: measured values (0.07–0.18 $\text{W m}^{-1} \text{K}^{-1}$) fall between our point-contact lower bound (0.022–0.031) and a continuous-skeleton parallel+Knudsen upper bound (0.080–0.155), close to the latter — as expected for a stretched-film, co-continuous polymer skeleton (Bruggeman, co-continuous: 0.090, in-range). A continuum sphere-pack model reproduces the same numbers (0.075–0.093) only by cancelling overestimated gas

Table 1: Baselines at the uncalendered points (dry, He pore gas with Knudsen correction, identical constituent inputs; coating values derived from [?] via Eq. (??)). All in $\text{W m}^{-1} \text{K}^{-1}$.

	measured	Wiener ⁻ (series)	Maxwell-Eucken	Bruggeman	Wiener ⁺ (parallel)	M0 (ZBS+K)
graphite (thin), $\psi = 0.606$	1.429	0.242	0.430	8.07	31.6	0.871
graphite (thick), $\psi = 0.597$	1.405	0.245	0.441	9.07	32.3	0.903
NMC622, $\psi = 0.513$	0.704	0.258	0.434	0.825	1.29	0.498
NMC811, $\psi = 0.507$	0.492	0.260	0.440	0.841	1.30	0.505

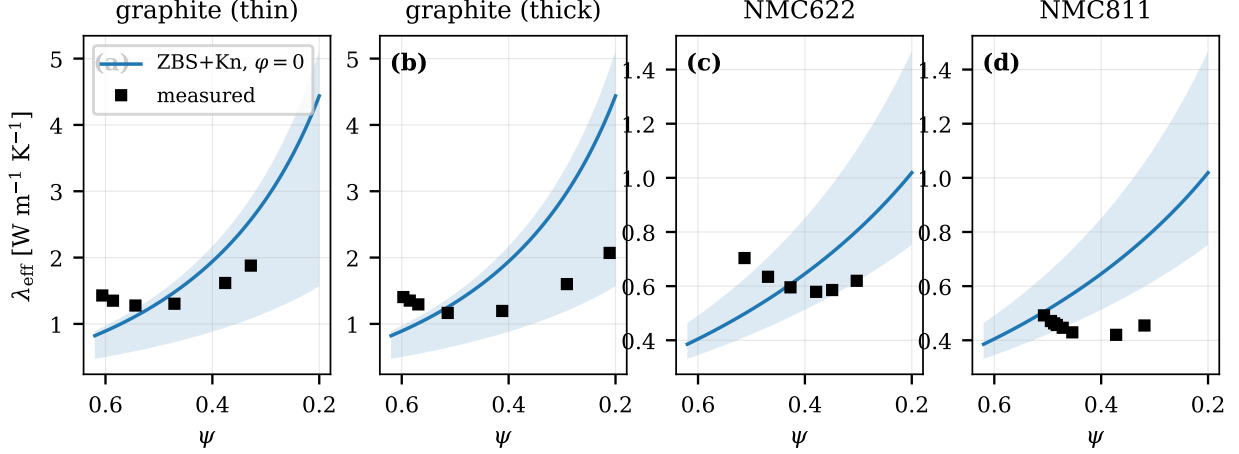


Figure 3: Zero-fit M0 (dry, He) against the coating conductivities of all 27 calendering states: (a) thin and (b) thick graphite anode, (c) NMC622, (d) NMC811. Bands: λ_s across the physical anisotropy range. Residuals grow systematically toward low porosity — the contact signature.

conduction against underestimated skeleton connectivity; the two-pressure experiment of Sec. ?? breaks exactly this degeneracy.

5.3 Ablation: what each model component buys

Table ?? isolates the contributions. Adding a *constant* contact fraction (M1) more than halves the error ($31.1\% \rightarrow 13.5\%$): the as-coated bridge network is first-order physics. Making the contact fraction *process-dependent* (M2, Eq. (??)) reaches 4.5% — the dataset’s own $\approx 5\%$ scatter — and reproduces the u-shape (Fig. ??): the interplay of bridge damage ($a < 0$, all families) against densification, with recovery ($b > 0$) exactly in the two families where interlocking is independently evidenced (SEM-documented particle penetration into the soft Al foil for NMC622; the highest-line-load thick anode [?]). Fitted parameters are physically legible: $\varphi_0 = 0.005\text{--}0.017$ brackets the VDI rigid-sphere contact fraction 0.0077 [?], and fitted λ_s falls in the c-axis-dominated low end of graphite’s anisotropy band, as through-plane transport across flakes requires. Held-out transfer: within-recipe 20.9% (thin $\rightarrow$ thick graphite); cross-recipe 39.5% (NMC622 $\rightarrow$ NMC811) — dissected next.

Table 2: Ablation: MAPE over all calendering states. M0: $\varphi = 0$, mid-band λ_s . M1: constant φ , fitted (λ_s, φ_0). M2: full $\varphi(\Pi)$, fitted ($\lambda_s, \varphi_0, a, b$). Right: M2 parameters.

Family	M0	M1	M2	M2 parameters ($\lambda_s, \varphi_0, a, b$)
graphite (thin)	27.3%	9.7%	1.8%	(24.8, 0.0094, -0.024 , 0)
graphite (thick)	48.6%	17.0%	5.4%	(5.0, 0.0121, -0.042 , 0.053)
NMC622	19.1%	14.2%	1.4%	(1.6, 0.0171, -0.089 , 0.039)
NMC811	29.3%	12.9%	9.4%	(1.5, 0.0048, -0.049 , 0)
average	31.1%	13.5%	4.5%	—

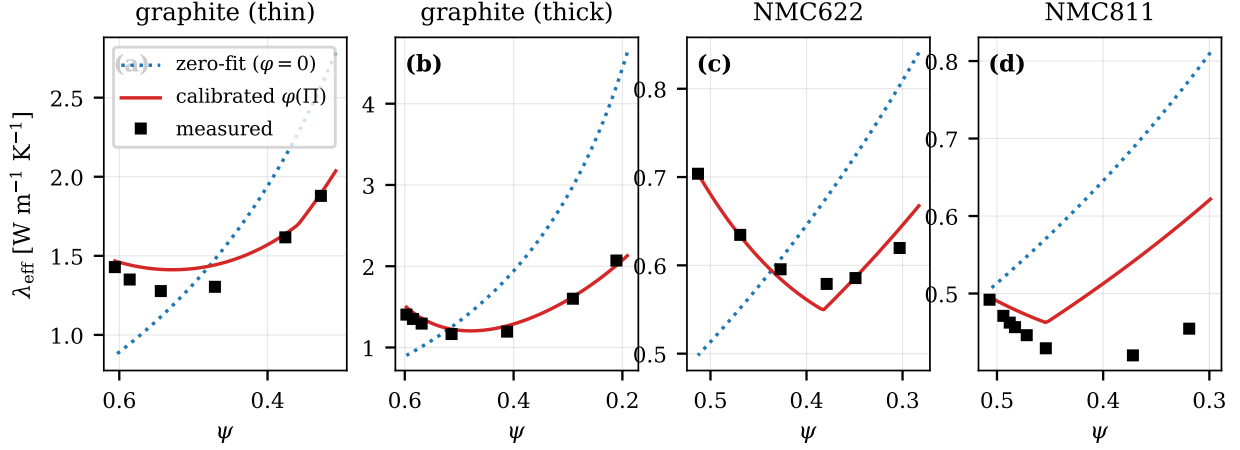


Figure 4: Calibrated M2 closure (solid) vs. M0 (dotted) along each family's calendaring trajectory; squares: measured coating values. The process-dependent contact term reproduces the u-shape that porosity-only closures cannot.

Table 3: Volume-space decomposition of the contact parameters.

Family	binder+additive [vol% of solid]	$G = \varphi_0 \lambda_b$ [W m ⁻¹ K ⁻¹]	G/vol	$ a /\varphi_0$ [Π^{-1}]	binder
graphite (thin)	6.4	1.22	19.0	2.6	CMC/SBR
graphite (thick)	6.4	1.57	24.4	3.5	CMC/SBR
NMC622	17.8	0.41	2.3	5.2	PVDF
NMC811	10.0	0.12	1.2	10.2	PVDF

5.4 Design principles from the fitted parameters

Because binder and additive densities (1.6–1.8 g cm⁻³) are far below the active materials' (NMC 4.7), recipes must be compared in *volume* space; Table ?? then decomposes the fits into three material interplays. (i) *Bridge conductance*. The as-coated group $G = \varphi_0 \lambda_b$ per unit inactive volume is $\approx 20\times$ larger for the anodes than the cathodes: soft graphite flakes deform plastically and self-bridge with high-conductivity contacts, whereas hard NMC particles depend entirely on the additive network — within which 2 wt% of flake graphite doubles NMC622's effectiveness over carbon-black-only NMC811. (ii) *Damage rate*. The fractional loss $|a|/\varphi_0$ orders perfectly by binder mechanics: stiff PVDF networks fail 2–4 $\times$ faster per unit compression than elastomeric CMC/SBR. (iii) *Recovery* is a process threshold, not a material constant: NMC811's $b = 0$ reflects a regime never reached ($\Pi \leq 0.26$), not absent physics. Design rules follow directly: a small flake-graphite additive outperforms additional carbon black ($\approx 2\times$ per unit volume); elastomeric co-binders protect the thermal/adhesion network in heavily calendered designs; and the calendaring setpoint should avoid the conductivity/adhesion dip — calender lighter, or push through to interlocking.

5.5 Error analysis: identifiability and transfer

Identifiability. Figure ??a profiles the likelihood over λ_s : fixing λ_s anywhere in [5, 30] W m⁻¹ K⁻¹ and refitting keeps MAPE within 2 points of optimum, with φ_0 compensating between 0.0073 and 0.0120; beyond $\lambda_s \approx 40$ the fit degrades irrecoverably. The data thus *do* reject the high (in-plane-dominated) end of the anisotropy band, but inside the valley individual parameters trade off — so the robust deliverables are the grouped quantities of Table ?? and the valley bounds as honest parameter uncertainty.

Transfer decomposition. Figure ??b explains the cross-recipe failure by hybrid swaps: applying NMC622's calibration to NMC811 yields 39.5%; replacing only λ_s recovers almost nothing (36.2%); replacing only φ_0 recovers $\approx 90\%$ of the recoverable error (12.2% vs. the 9.4% self-fit floor); adding (a, b) contributes a further 0.3 points. Cross-recipe transfer failure is therefore *recipe physics concentrated in one parameter* — the as-coated bridge conductance set by the additive system — with the practical corollary that one as-coated measurement re-anchors φ_0 when changing recipes, while the damage dynamics transfer within a chemistry.

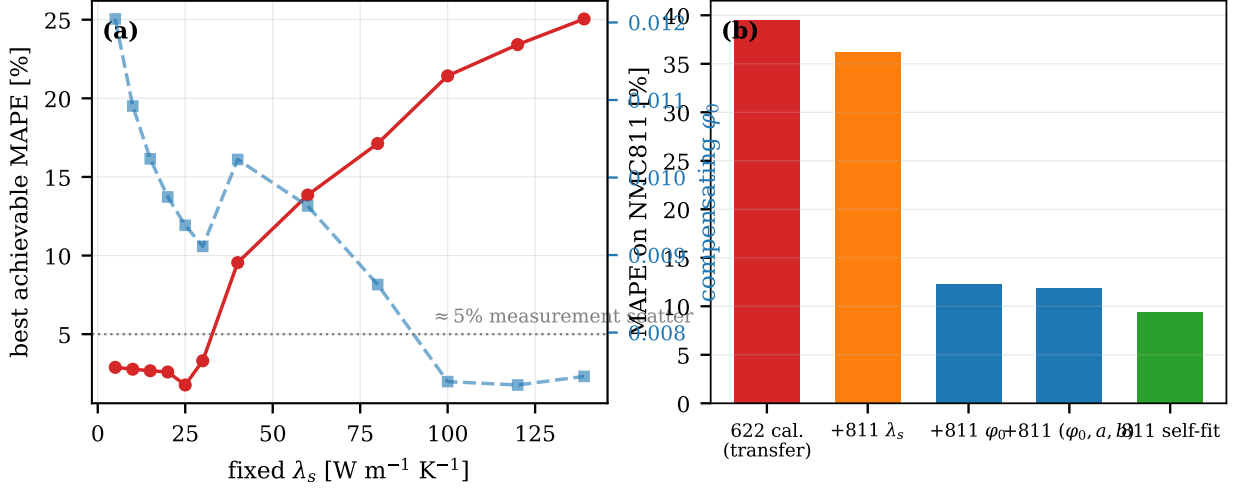


Figure 5: (a) Profile likelihood over λ_s (thin anode): the λ_s/ϕ_0 identifiability valley; right axis, compensating ϕ_0 . (b) Hybrid-swap decomposition of the NMC622→NMC811 transfer: the as-coated bridge term ϕ_0 carries the failure.

Remaining exposures. The Knudsen coefficient $\beta \in [1.5, 2.0]$ moves dry electrode predictions by $\leq 1.7\%$ but the dry separator by 11.7% (resolved by the two-pressure experiment); 6 of 27 sheets lie below the packed-bed-validated porosity range with no degradation of calibrated residuals (the extrapolation is absorbed by $\varphi(\text{II})$, empirically); NMC811’s recovery term must not be extrapolated beyond its tested compression; and all coating values inherit the lumped coating/collector contact resistance of stack metrology [?].

5.6 Manufacturing QC applications

Porosity QC. On a synthetic batch of 200 calendered sheets ($\psi \sim \mathcal{N}(0.30, 0.012)$; 3% measurement noise), Newton inversion (Eq. (??)) recovers porosity with RMSE 0.0078 absolute and zero bias (Fig. ??); the a-priori prediction of Eq. (??), $\sigma_\psi = 0.0092$ including the 2% model floor, brackets the observed scatter — the uncertainty model is calibrated, slightly conservative. Against a ± 0.02 spec window this yields 87.5% single-shot classification; $n = 4$ averaging makes the window cleanly resolvable.

Interface/delamination QC. The calibrated closure decomposes λ_{eff} into core and bridge contributions. As-coated — precisely where post-drying adhesion and binder-migration QC matter — bridges carry 16–66% of the heat, so a bridge-network failure shifts a thermal reading by $z = 5$ –22 standard deviations at 3% noise: even partial degradation is single-shot detectable. Porosity is already measured inline (thickness and mass loading); *interface quality is not* — on this evidence, a fast thermal reading is its most direct inline proxy, consistent with the independently reported conductivity–adhesion correlation [?].

6 Discussion: choosing among approaches

The results support a clear division of labor. *Analytic closures* remain the right tool whenever evaluations must be cheap, differentiable, or inverted — cell-level thermal simulation, process optimization, QC — provided two conditions hold: the morphology assumption matches the material (Sec. ?? shows baseline *failure patterns* themselves identify the morphology: point-contact for low-additive electrodes, co-continuous for separators), and process-dependent contact physics is represented (Table ??; it is worth 31.1% → 4.5%). *Microstructure-resolved simulation* is the natural tool for exploring hypothetical microstructures before they exist, but at present neither DEM nor homogenization generators encode calender-induced contact evolution [?]; embedding a $\varphi(\text{II})$ -like contact model into those generators is an obvious synthesis. *Purely data-driven regression* would fit the 27 points trivially but offers no mechanism, no transfer diagnosis, and no pressure- or gas-dependence to test; the closure’s value lies in its parameters *meaning* something — which is what turns a fit into design rules (Table ??) and into falsifiable predictions (Sec. ??). The per-recipe nature of the contact parameters, quantified by the transfer analysis, is then not a defect but a measurement: contact conduction lives in the contact zones, which bulk volume averages cannot see; six

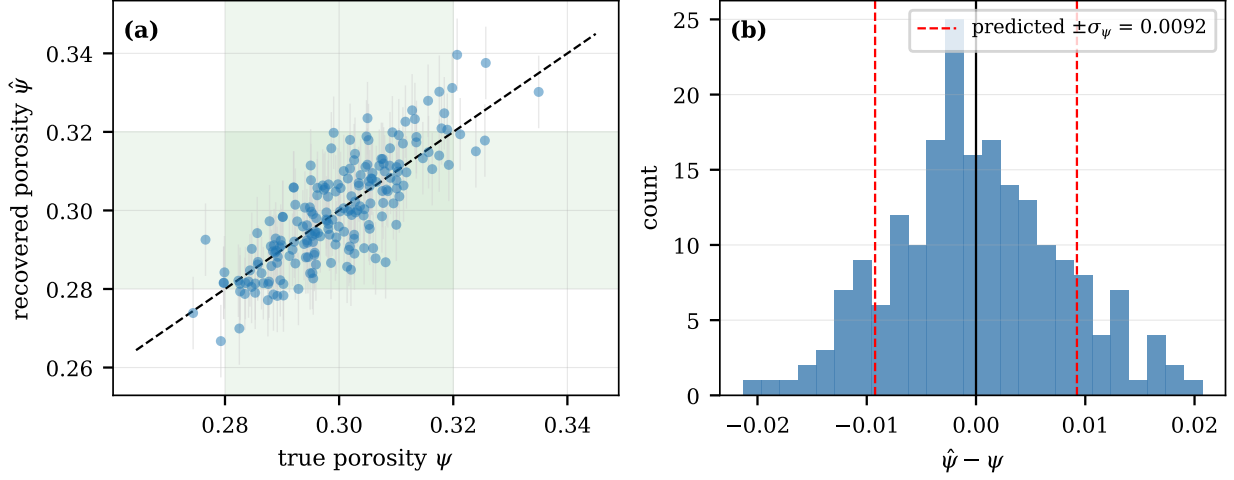


Figure 6: Inverse porosity QC, 200-sheet synthetic batch at 3% noise: (a) parity with predicted $\pm\sigma_\psi$ and ± 0.02 spec window; (b) recovery-error histogram vs. the first-order uncertainty prediction (Eq. (??)).

calendering states suffice to pin it for a new recipe, and a single as-coated point re-anchors the dominant parameter.

7 Proposed experimental validation

The model is calibrated on published data; the following campaign (per recipe: ≈ 26 sheets, one calendering afternoon, two measurement days) tests its *mechanisms*.

1. **Sample matrix.** Six calendering states from $\Pi = 0$ to the production maximum (single batch, single-side coatings); duplicate as-coated sheets for the φ_0 re-anchor; each state dry and electrolyte-soaked.
2. **Porosity ground truth.** Areal mass; flat-anvil micrometer thickness ($\leq 0.3 \text{ N mm}^{-2}$); helium-pycnometric solid density; GUM uncertainty budget.
3. **Conductivity.** Through-plane LFA (penetration model [?]; purge gas recorded — it is the pore gas) or guarded hot plate; 20°C ; $n = 4$ repeats ($3\% \rightarrow 1.5\%$ effective noise).
4. **Two-pressure Knudsen discrimination** (decisive for separators; recommended for the most porous electrode state): repeat dry measurements at ≈ 1000 and ≤ 100 mbar. By Eq. (??), pore-gas conduction depends on pressure through $\text{Kn} \propto 1/p$; skeleton and bridge conduction do not. The pressure split isolates the gas channel, pins β , and directly breaks the error cancellation identified for continuum sphere-pack separator models (Sec. ??).
5. **Adhesion correlation.** Pull-off strength at every calendering state: Eq. (??) predicts the conductivity-relevant contact fraction tracks interface damage and interlocking, so $\varphi(\Pi)$ and adhesion must share sign structure and dip location.
6. **Acceptance criteria.** (i) Calibrated residuals $\leq$ measurement uncertainty at all Π ; (ii) pressure split consistent with Eq. (??) within the β band; (iii) $\varphi(\Pi)$ –adhesion correlation as predicted. Failure of (ii) falsifies the Knudsen term as implemented; failure of (iii) falsifies the contact interpretation, leaving $\varphi(\Pi)$ as curve-fit only.

8 Limitations

- (1) All conclusions about the contact term rest on *one* (excellent) public dataset — four families, three recipes, one lab, one metrology chain; the campaign of Sec. ?? is required before generality is claimed.
- (2) With 4 parameters on 6–8 points per family, λ_s and φ_0 are confounded at $\Pi = 0$; we report the profile-likelihood valley and grouped quantities rather than over-interpreting point estimates.
- (3) The quadratic $\varphi(\Pi)$ is phenomenology anchored to adhesion observations, not derived contact mechanics; b is unconstrained outside each family’s tested compression range.
- (4) Stack-derived coating values lump the coating/collector contact resistance; single-coating metrology would separate it.
- (5) Electrode porosities

extrapolate the packed-bed-validated porosity range of the base closure (handled empirically, not mechanically). (6) The sphere-pack morphology is wrong for stretched-film separators; we therefore claim only bounds there, not point predictions. (7) Anisotropy is compressed into a scalar effective λ_s ; in-plane conduction and anisotropic unit cells are out of scope. (8) Wet-state predictions assume full pore saturation; partial wetting during electrolyte filling is untreated.

9 Conclusion

A minimal, physically interpretable extension of the ZBS closure — Knudsen-corrected pore gas plus a compression-indexed contact term — closes the gap between packed-bed theory and calendered battery electrodes: it quantifies the rarefaction correction the literature had only flagged, outperforms the classical effective-medium hierarchy zero-fit, reaches the measurement noise floor once calibrated (4.5% over 27 calendering states), and is the first closure to reproduce the measured u-shaped conductivity–calendering dependence. Its parameters are not nuisance constants but measurements of contact physics: they decompose into design rules (additive choice, binder mechanics, calendering setpoint), their identifiability is bounded by profile likelihood, and their transfer behavior localizes recipe dependence in a single re-anchorable number. Differentiability turns the same closure into manufacturing instrumentation — porosity to ± 0.008 from one thermal reading, and the first candidate inline monitor for interface quality. The specified two-pressure campaign will test the mechanisms, not merely the fits.

Reproducibility statement

All results derive from executed code in the accompanying repository: the closure implementation (JAX, 64-bit; $\varphi = 0$ reduction to the validated base closure is regression-tested), dataset transcriptions with per-value citations, calibration scripts with stated objectives, bounds and seeds (Sec. ??), and a figure pipeline that regenerates every figure from raw inputs with no hard-coded numbers. A 14-test suite locks the headline values (including the NMC811 +2.7% prediction) against silent drift. Random seeds: QC batch `default_rng(42)`; all optimizer settings in the text. [Repository URL/DOI to be added.]

Data availability

The transcribed datasets (`gandert2023_calendering.csv`, `separator_electrode_literature.csv`) carry row-level citations to the primary sources [?, ?, ?, ?, ?]; the primary calendering dataset is published open access by its authors.

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[To be completed — colleagues, institutional support, funding.]

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